

Data Representation

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Data Representation

p_{data}



$$\mathbf{x}^j \sim p_{data}$$
$$j = 1, 2, \dots, |\mathcal{D}|$$

- dataset $\mathcal{D}$
- data distribution p_{data}
- model parameters $\theta \in \mathcal{M}$

- **How to represent (model) a data distribution?**

It can be an optimisation problem:

$$\min_{\theta \in \mathcal{M}} \mathcal{L}(p_{data}, p_{\theta})$$

- **Why parametric models?**

They scale more efficiently with large dataset than non-parametric models.

Data Representation

p_{data}



$$\mathbf{x}^j \sim p_{data}$$
$$j = 1, 2, \dots |\mathcal{D}|$$

- dataset $\mathcal{D}$
- data distribution p_{data}
- model parameters $\theta \in \mathcal{M}$

- We want to learn a probability distribution $p(\mathbf{x})$ over $\mathbf{x}$
1. **Generation (sampling):** $\mathbf{x}_{new} \sim p(\mathbf{x})$
 2. **Density Estimation:** $p(\mathbf{x})$ high if $\mathbf{x}$ looks like a cat
 3. **Unsupervised Representation Learning:**
Discovering the underlying structure from the data distribution (e.g., ears, nose, eyes ...)

Data Representation

- Recap: Challenges from Lecture 1

- Representation ability

How to represent $p(x)$

For 1-D data x , the probability distribution $p(x)$ is simple, e.g., Gaussian?

For high-dimensional data $\mathbf{x} = (x_1, x_2, \dots, x_n)$,

how do we learn the joint distribution $p(x_1, x_2, \dots, x_n)$?

- Learning method

How do we measure and minimise the distance

between the estimated distribution $p(x)$ (i.e., p_x) and the real distribution p_{data} ?

we can now perform generative process and density estimation

- Inference

How do we perform discriminative task?

i.e., invert the generative process

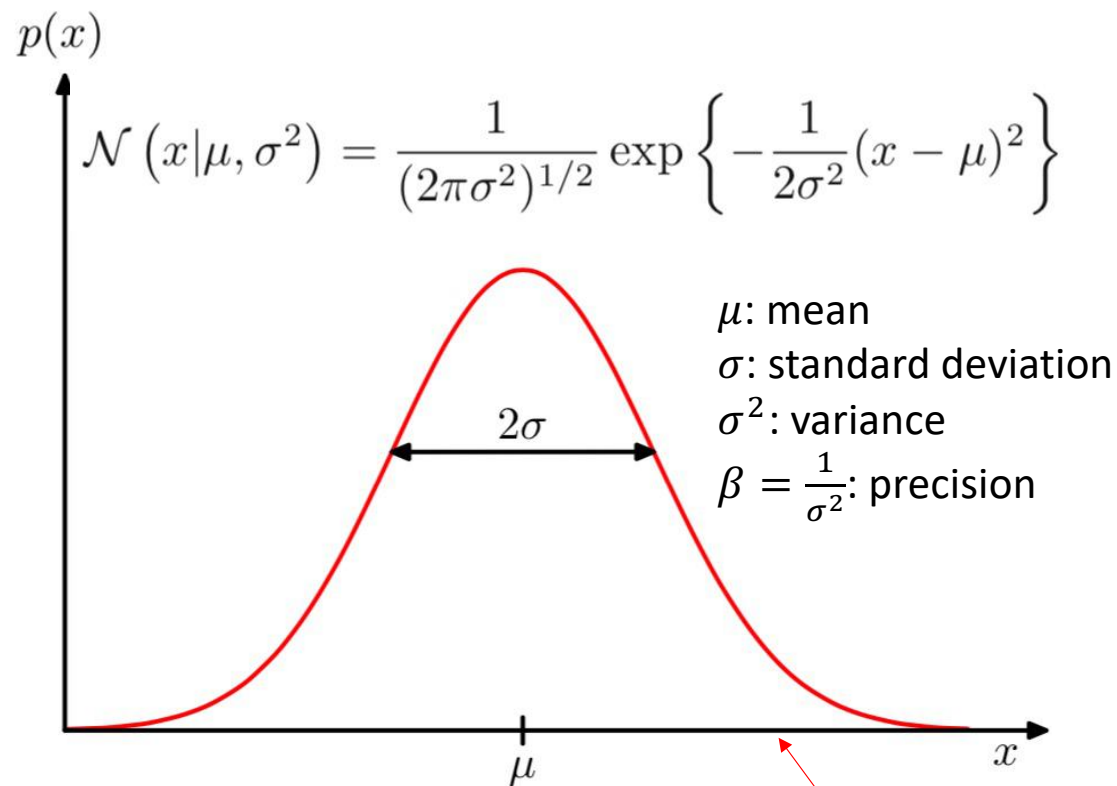
Data Representation

- How to represent data
 - Problem of High-dimensional Data
 - Less Parameters: Conditional Independence
 - Less Parameters: Bayesian Network
- How to inference
 - Naïve Bayes Classifier
 - Discriminative vs. Generative Models
 - Logistic Regression
- How to be better
 - Deep Neural Networks
 - Continuous Variables

- Problem of High-dimensional Data
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Problem of High-dimensional Data

- How to represent the height distribution (age from 30cm to 230cm)



The probability of x to be this value

height

In this case, we have 200 states
and we need 2 parameters to represent
the probability distribution $p(x)$

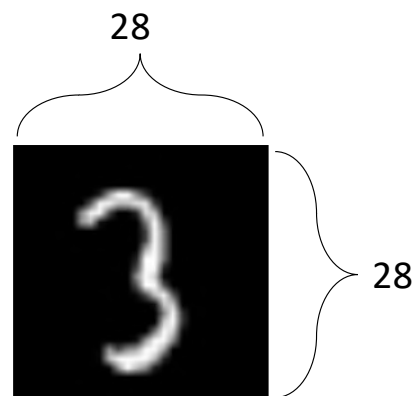
μ, σ

Problem of High-dimensional Data

- How to represent a **high-dimensional data** $\mathbf{x} = (x_1, x_2, x_3, \dots, x_n)$



MNIST dataset



784 random binary variables

In MNIST, an images have $28 * 28 * 1 = 784$ binary values

So... how to represent $p(x_1, x_2, \dots, x_{784})$?
how many number of parameters?

Problem of High-dimensional Data

- How to represent a **high-dimensional data** $\mathbf{x} = (x_1, x_2, x_3, \dots, x_n)$

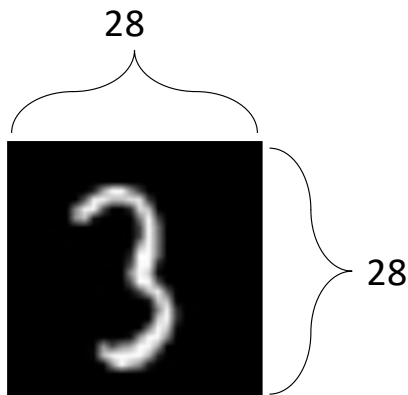
(Bernoulli random variables)

As x can be either 0 or 1, i.e., only 2 states

(Joint distribution)

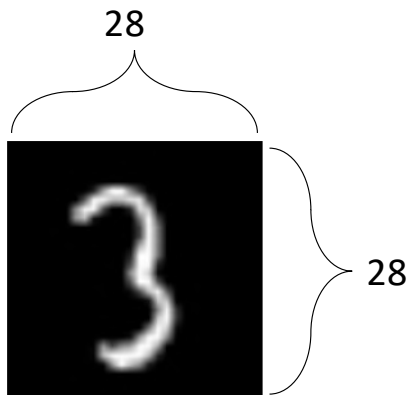
The number of possible state for $p(x_1, x_2, \dots, x_n)$ is 2^n
which is far larger than the number of data sample

We need a super-large memory to store $p(x_1, x_2, \dots, x_n)$
even we have such large memory, we do not have enough data
to learn/model it



Problem of High-dimensional Data

- How to represent a **high-dimensional data** $\mathbf{x} = (x_1, x_2, x_3, \dots, x_n)$



784 random binary variables

$p(x_1, x_2, \dots, x_n)$ has 2^n states, then ...

How many number of parameters to model $p(x_1, x_2, \dots, x_n)$? : $2^n - 1$?

Recap: **Product Rule**

$$p(x_1, x_2) = p(x_1)p(x_2|x_1)$$

$$p(x_1, x_2, x_3) = p(x_1, x_2)p(x_3|x_1, x_2) = p(x_1)p(x_2|x_1)p(x_3|x_1, x_2)$$

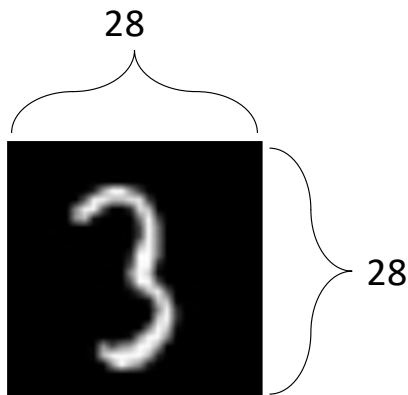
...

$$p(x_1, x_2, \dots, x_n) = p(x_1)p(x_2|x_1)p(x_3|x_1, x_2) \dots p(x_n|x_1, \dots, x_{n-1})$$

Problem of High-dimensional Data

- How to represent a **high-dimensional data** $\mathbf{x} = (x_1, x_2, x_3, \dots, x_n)$

$$p(x_1, x_2, \dots, x_n) = p(x_1)p(x_2|x_1)p(x_3|x_1, x_2) \dots p(x_n|x_1, \dots, x_{n-1})$$



784 random binary variables

- $p(x_1)$ need 1 parameter, the probability of x_1 to be 1 (as it is a binary variable)
- $p(x_2|x_1)$ need 2 parameters, i.e., $p(x_2|x_1 = 0)$ and $p(x_2|x_1 = 1)$
- $p(x_3|x_1, x_2)$ need 4 parameters, i.e., $p(x_3|x_1 = 0, x_2 = 0)$, $p(x_3|x_1 = 0, x_2 = 1)$, $p(x_3|x_1 = 1, x_2 = 0)$, $p(x_3|x_1 = 1, x_2 = 1)$

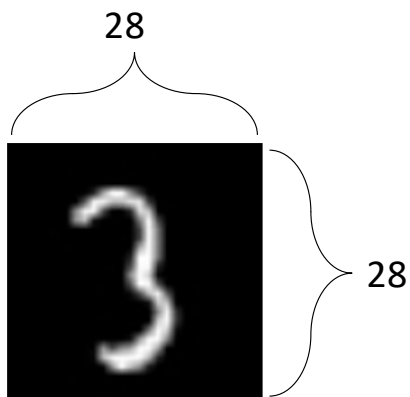
So ... The number of parameters to model $p(x_1, x_2, \dots, x_n)$ is:

$$1 + 2 + 4 + \dots + 2^{n-1} = 2^n - 1$$

(when variables are binary)

Problem of High-dimensional Data

- How to represent a **high-dimensional data** $\mathbf{x} = (x_1, x_2, x_3, \dots, x_n)$



784 random binary variables

Product Rule:

$$p(x_1, x_2, \dots, x_n) = p(x_1) p(x_2|x_1) p(x_3|x_1, x_2) \dots p(x_n|x_1, \dots, x_{n-1})$$

2^n states $2^n - 1$ parameters

$2^n - 1$ is exponential,
the product rule does not help to reduce the num of parameters

Problem of High-dimensional Data

- How to represent a **high-dimensional data** $\mathbf{x} = (x_1, x_2, x_3, \dots, x_n)$

In practice

- 1) The x can be continuous, i.e., infinite states
- 2) The number of x can be millions

For simplicity

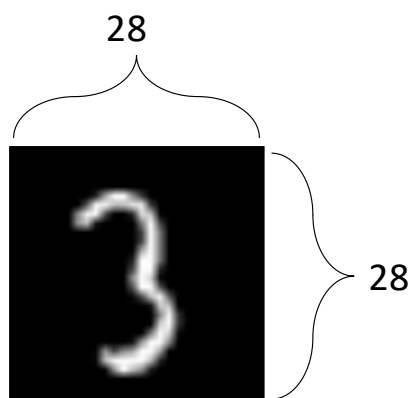
We use binary x and MNIST for demo

- Problem of High-dimensional Data
- **Less Parameters: Conditional Independence**
- Less Parameters: Bayesian Network
- Naïve Bayes Classifier
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Less Parameters: Conditional Independence

- How to reduce the number of parameter to represent $p(x_1, x_2, \dots, x_n)$?

Product Rule does not help:



$$p(x_1, x_2, \dots, x_n) = p(x_1) p(x_2 | x_1) p(x_3 | x_1, x_2) \dots p(x_n | x_1, \dots, x_{n-1})$$

2^n states $2^n - 1$ parameters

784 random binary variables

Less Parameters: Conditional Independence

- How to reduce the number of parameter to represent $p(x_1, x_2, \dots, x_n)$?

Recap: If variables x_1, x_2 are conditional independent given variable x_3 ,
denotes as $x_1 \perp x_2 \mid x_3$

$$p(x_1, x_2 \mid x_3) = p(x_1 \mid x_3)p(x_2 \mid x_3)$$

if not independent:

$$p(x_1, x_2 \mid x_3) = \frac{p(x_1, x_2, x_3)}{p(x_3)} = \frac{p(x_1, x_2, x_3)}{p(x_2, x_3)} \frac{p(x_2, x_3)}{p(x_3)} = p(x_1 \mid x_2, x_3)p(x_2 \mid x_3)$$

so we can have $p(x_1 \mid x_2, x_3)p(x_2 \mid x_3) = p(x_1 \mid x_3)p(x_2 \mid x_3)$ **if** $x_1 \perp x_2 \mid x_3$

Less Parameters: Conditional Independence

- How to reduce the number of parameter to represent $p(x_1, x_2, \dots, x_n)$?

Given product rule: $p(x_1, x_2, x_3, x_4) = p(x_1)p(x_2|x_1)p(x_3|x_1, x_2)p(x_4|x_1, x_2, x_3)$

If $x_4 \perp x_2 \mid \{x_1, x_3\}$, we can simplify it as:

$$p(x_1, x_2, x_3, x_4) = p(x_1)p(x_2|x_1)p(x_3|x_1, x_2)p(x_4|x_1, \textcolor{red}{x_2}, x_3)$$

If $x_2 \perp \{x_1, x_3\} \mid x_4$, we can simplify it as:

$$\begin{aligned} p(x_1, x_2, x_3, x_4) &= p(x_4, x_3, x_2, x_1) = p(x_4)p(x_3|x_4)p(x_2|x_3, x_4)p(x_1|x_2, x_3, x_4) \\ &= p(x_4)p(x_3|x_4)p(x_2|x_3, x_4)p(x_1|\textcolor{red}{x_2}, \textcolor{red}{x_3}, x_4) \end{aligned}$$

Less Parameters: Conditional Independence

- How to reduce the number of parameter to represent $p(x_1, x_2, \dots, x_n)$?

In an **extreme case**, if $x_{i+1} \perp \{x_1, x_2 \dots x_{i-1}\} \mid x_i$, i.e., the next variable only related to the current variable (Markov model!)

$$\begin{aligned} p(x_1, x_2, x_3, x_4) &= p(x_1)p(x_2|x_1)p(x_3|x_1, x_2)p(x_4|x_1, x_2, x_3) \\ &= p(x_1)p(x_2|x_1)p(x_3|x_1, x_2)p(x_4|x_1, x_2, x_3) \\ &= p(x_1)p(x_2|x_1)p(x_3|x_2)p(x_4|x_3) \end{aligned}$$

If x are binary variables

$$2n - 1 \text{ parameters} \ll 2^n - 1 \text{ parameters}$$

So ...

if conditional independencies exist, the number of parameter can be reduced!!

Less Parameters: Conditional Independence

- How to reduce the number of parameter to represent $p(x_1, x_2, \dots, x_n)$?

In a **MORE extreme case**, if x_i are independent identical (IID)

$$\begin{aligned} p(x_1, x_2, x_3, x_4) &= p(x_1)p(x_2|x_1)p(x_3|x_1, x_2)p(x_4|x_1, x_2, x_3) \\ &= p(x_1)p(x_2|x_1)p(x_3|x_1, x_2)p(x_4|x_1, x_2, x_3) \\ &= p(x_1)p(x_2)p(x_3)p(x_4) \end{aligned}$$

However, in practice, there exists “**relationship**” between variables
the independence assumption is not practical...

e.g., the following random samples would not happen



- Problem of High-dimensional Data
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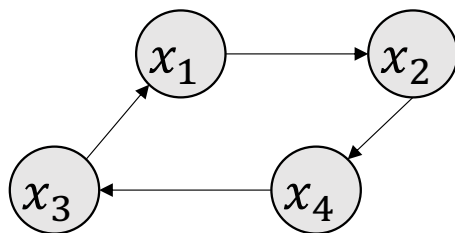
Less Parameters: Bayesian Network

- Key idea:

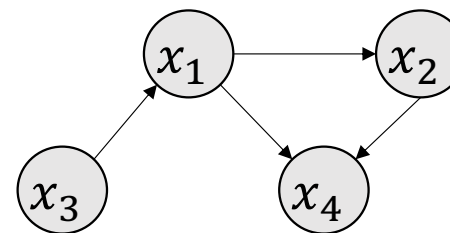
$$\text{Joint distribution: } p(x_1, x_2, \dots, x_n) = p(x_1) \underbrace{p(x_2|x_1)p(x_3|x_1, x_2) \dots p(x_n|x_1, \dots, x_{n-1})}_{2^n - 1 \text{ parameters if } x \text{ are binary variables}}$$

use conditional distribution instead of joint distribution to reduce the num of parameters

Bayesian network structure is a **Directed Acyclic Graph**, $G = (V, E)$
where V means vertexes, E means edges



Directed Cycle



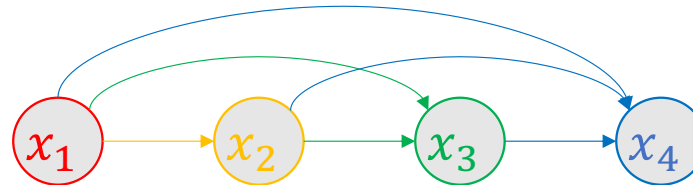
Directed Acyclic Graph

Less Parameters: Bayesian Network

- Key idea:

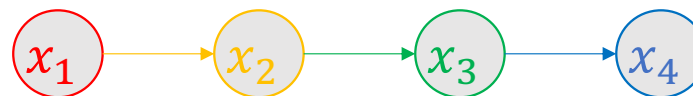
Bayesian network structure is a **Directed Acyclic Graph**, $G = (V, E)$

Joint distribution: $p(x_1, x_2, x_3, x_4) = p(x_1)p(x_2|x_1)p(x_3|x_1, x_2)p(x_4|x_1, x_2, x_3)$



If $x_{i+1} \perp \{x_1, x_2 \dots x_{i-1}\} \mid x_i$

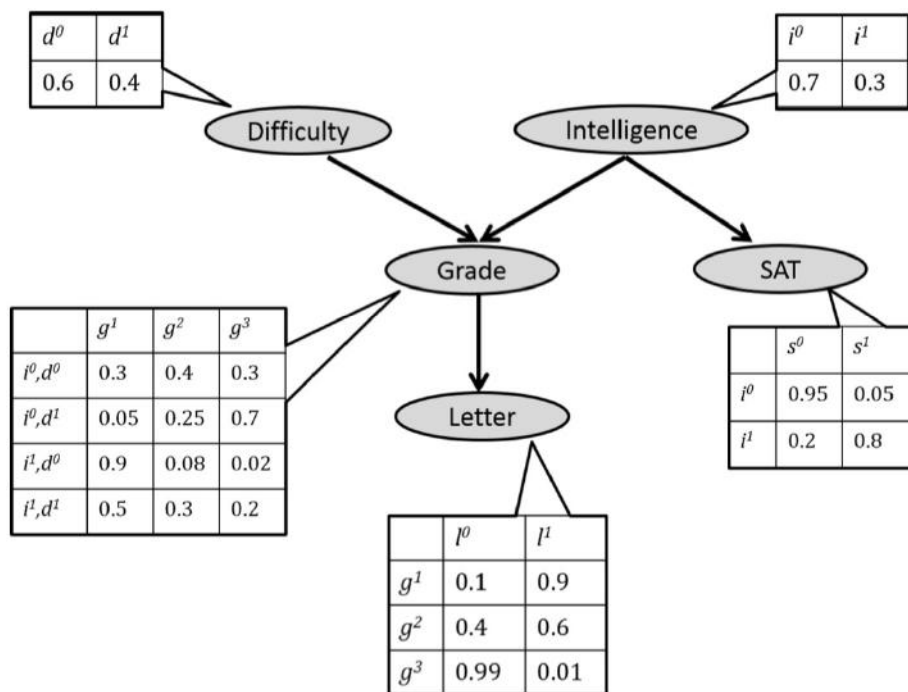
$$\begin{aligned}
 p(x_1, x_2, x_3, x_4) &= p(x_1)p(x_2|x_1)p(x_3|x_1, x_2)p(x_4|x_1, x_2, x_3) \\
 &= p(x_1)p(x_2|x_1)p(x_3|x_2)p(x_4|x_3)
 \end{aligned}$$



Less edges == Less parameters

Less Parameters: Bayesian Network

- Example



$$p(d, i, g, s, l) = p(d)p(i|d)p(g|d, i)p(s|d, i, g)p(l|d, i, g, s)$$

According to the left Bayesian Net, we have the independencies:

$$d \perp i \quad s \perp \{d, g\} \quad l \perp \{d, i, s\}$$

So that ..

$$p(d, i, g, s, l) = p(d)p(i)p(g|i, d)p(s|i)p(l|g)$$

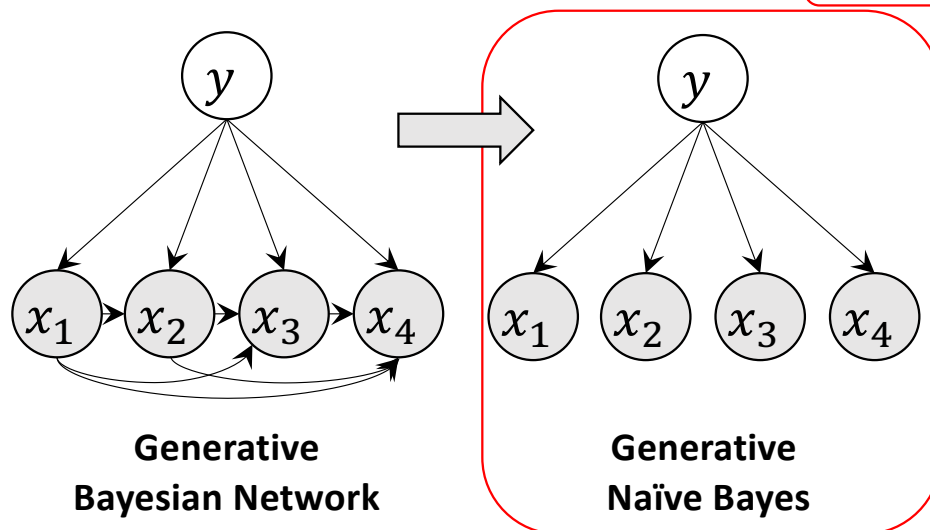
Less Parameters: Bayesian Network

- Bayesian Network structure is a **Directed Acyclic Graph**, $G = (V, E)$
- Bayesian Network is given by (G, P) ,
where P is a set of **local conditional probability distributions** for each node/vertex of G
- Compute the P using data samples to “learn” the Bayesian Network
- Bayesian Network is also known as **Belief Network** and **Bayes Network**

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Naïve Bayes Classifier

- How Bayesian Network performs inferencing? i.e., discriminative tasks?
- Suppose we have a binary classification problem, label $y = 0, 1$, features $\mathbf{x} = (x_1, x_2, x_3, x_4)$
- The probability distribution is $p(y, x_1, x_2, x_3, x_4)$
- Naïve Bayes Classifier assume that $x_i \perp \mathbf{x}_{-i} | y$, so that:



Given Naïve Bayes Assumption:

$$p(y, x_1, x_2, x_3, x_4) = p(y)p(x_1|y)p(x_2|y)p(x_3|y)p(x_4|y)$$

$$p(\mathbf{x}|y)$$

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we need $p(y|\mathbf{x})$

Naïve Bayes Classifier

$$p(y|\mathbf{x}) = \frac{p(\mathbf{x}, y)}{p(\mathbf{x})} = \frac{p(y)p(\mathbf{x}|y)}{p(\mathbf{x})} \propto p(y)p(\mathbf{x}|y)$$

$$\hat{y} = \arg \max_y p(y|\mathbf{x}) = \arg \max_y \frac{p(\mathbf{x}, y)}{p(\mathbf{x})} = \arg \max_y p(y)p(\mathbf{x}|y)$$

Given Naïve Bayes Assumption:

$$p(\mathbf{x}|y) = p(x_1|y)p(x_2|y)p(x_3|y)$$

Naïve Bayes Classifier

- Given $p(\mathbf{x}|y) = p(x_1|y)p(x_2|y)p(x_3|y)$, how to compute $p(Y|X)$?
- First, we can **estimate** the parameters from the training set:

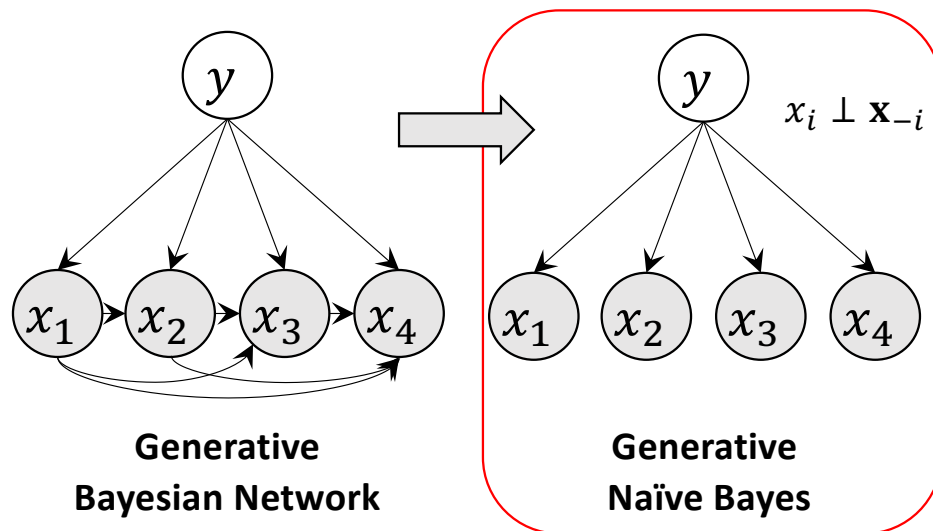
	$x_1 = 0$	$x_1 = 1$	$x_2 = 0$	$x_2 = 1$	$x_3 = 0$	$x_3 = 1$	$x_4 = 0$	$x_4 = 1$
$y = 0$	3	5	5	2	0	8	7	4
$y = 1$	1	0	3	10	7	4	2	5

- $$p(Y = 0) = \frac{3+5+5+2+8+7+4}{(3+5+5+2+8+7+4)+(1+3+10+7+4+2+5)}$$
- $$p(x_1 = 0|Y = 0) = \frac{3}{3+5+5+2+8+7+4}$$
-
- Second, **predict** the probability of a label given an input with **Bayes rule**:
 - $$p(Y = 0|x_1, x_2, x_3, x_4) = \frac{p(Y=0) \prod_{i=1}^4 p(x_i|Y=0)}{\sum_{y=\{0,1\}} p(Y=y) \prod_{i=1}^4 p(x_i|Y=y)}$$

Naïve Bayes Classifier

- Limitation

Are the independence assumptions reasonable ??



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Discriminative vs. Generative Models

← symmetry property $p(X, Y) = p(Y, X)$

- Given $p(Y, X) = p(X|Y)p(Y) = p(Y|X)p(X)$
- Discriminative: $X \rightarrow Y$, we only need to estimate the conditional distribution $P(Y|X)$ without learning to model $P(X)$
simply input X then output Y



- Generative: $Y \rightarrow X$, we need both $P(Y)$ and $P(X|Y)$ to compute $p(Y|X)$ via Bayes
(see the Naïve Bayes Classifier as an example)



$$p(Y|X) = p(X|Y)p(Y)/p(X)$$

Discriminative vs. Generative Models

- Given a random vector $\mathbf{x} = (x_1, x_2, \dots, x_n)$, the product rules can give us:

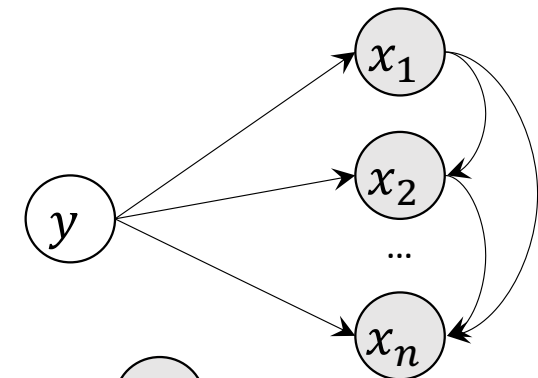
$$p(y, \mathbf{x}) = p(y)p(x_1|y)p(x_2|y, x_1) \dots p(x_n|y, x_1, x_2, \dots, x_{n-1})$$

$$p(y, \mathbf{x}) = p(x_1)p(x_2|x_1)p(x_3|x_1, x_2) \dots p(y|x_1, x_2, \dots, x_{n-1})$$

generative

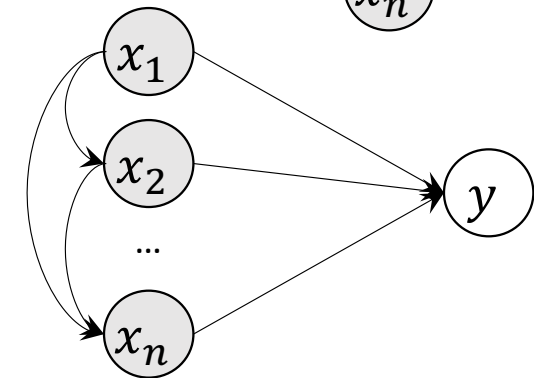
$p(y)$ is simple to estimate

but how to parametrise $p(x_i|y, x_1, \dots, x_{i-1})$?



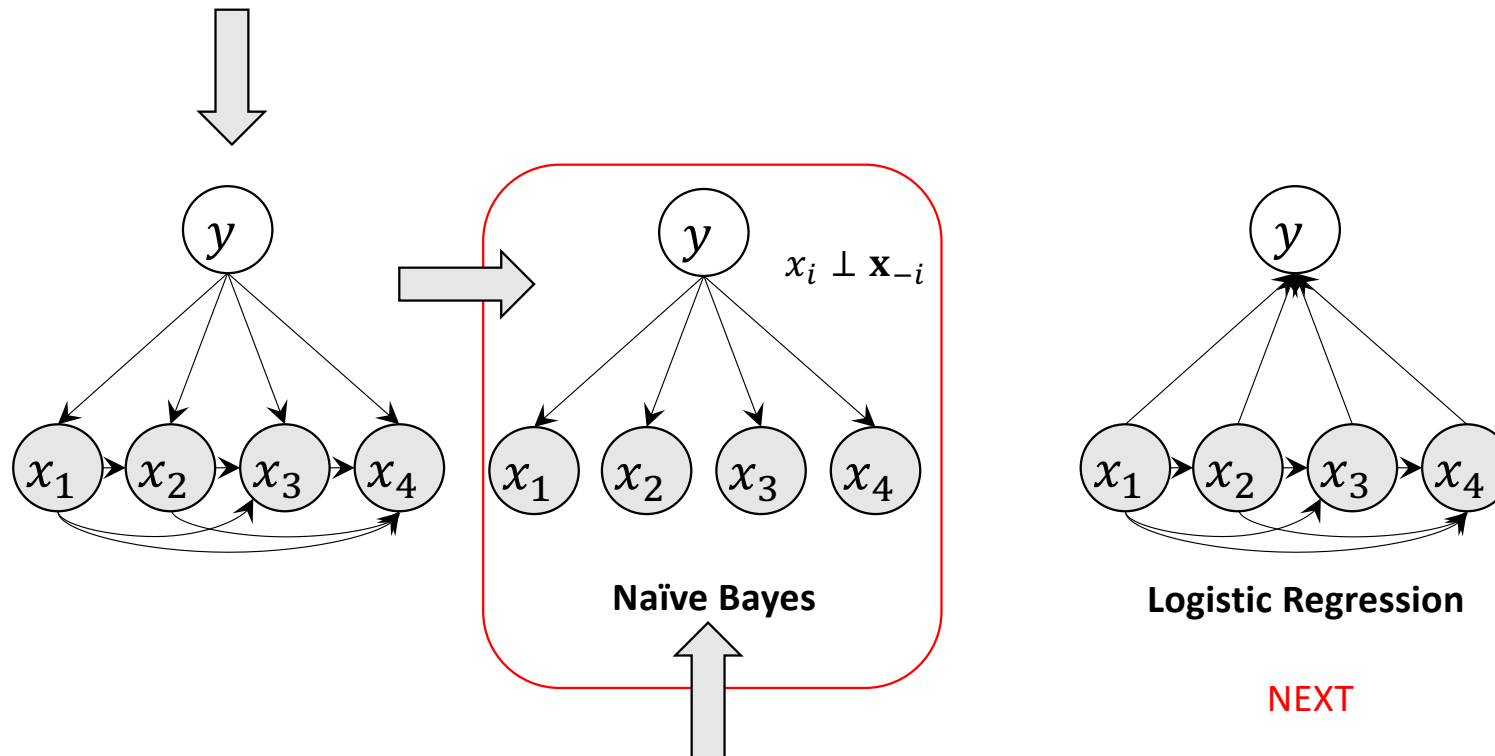
discriminative

only need to parametrise $p(y|x_1, \dots, x_{n-1})$



Discriminative vs. Generative Models

parametrise $p(x_i|y, x_1, \dots, x_{i-1})$ **without** independent assumptions



NEXT

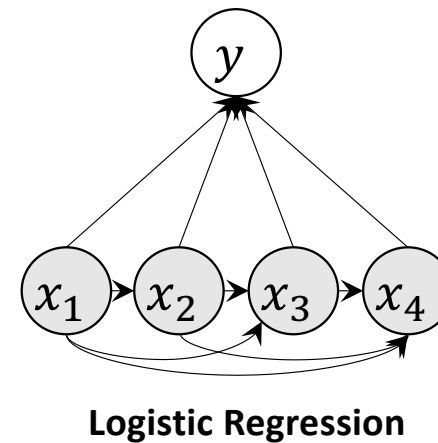
parametrise $p(x_i|y, x_1, \dots, x_{i-1})$ with independent assumptions

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Logistic Regression

- Parameterise the $p(Y|X)$ without independence assumptions

only need to parametrise $p(y|x_1, \dots, x_{n-1})$

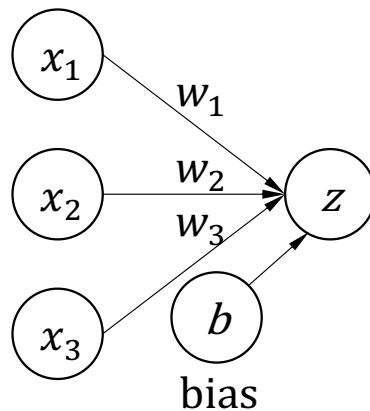


Logistic Regression

- (only need to) parameterise the $p(Y|X)$ without independence assumptions

$$p(Y = 1|\mathbf{x}, \mathbf{w}, b) = f(\mathbf{x}, \mathbf{w}, b)$$

input layer output layer



$$z = x_1w_1 + x_2w_2 + x_3w_3 + b$$

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad \mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix}$$

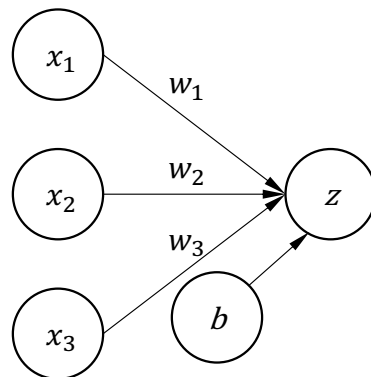
$$z = \mathbf{w}^T \mathbf{x} + b$$

$$z = [w_1 \quad w_2 \quad w_3] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + b$$

Logistic Regression

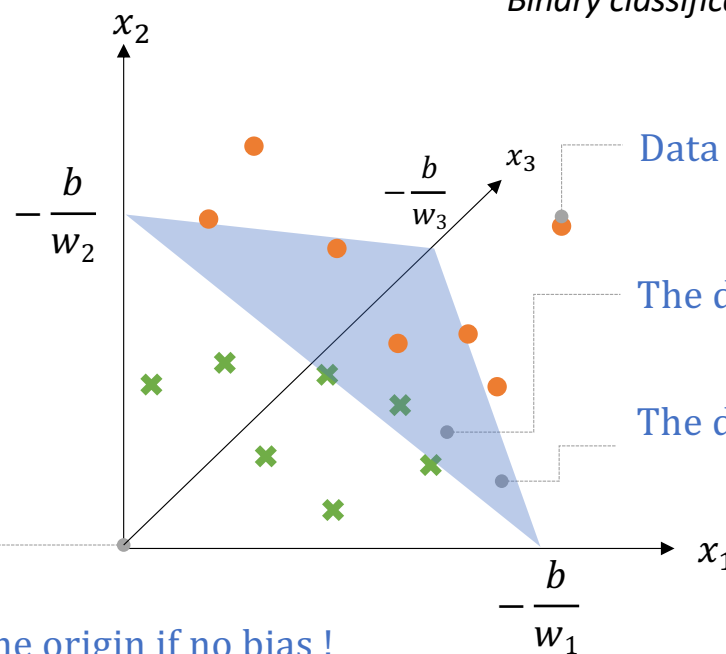
- (only need to) parameterise the $p(Y|X)$ without independence assumptions

input layer output layer



$$z = x_1w_1 + x_2w_2 + x_3w_3 + b$$

$$\text{Binary classification: } y = \begin{cases} 0, & \text{if } z \leq 0 \\ 1, & \text{if } z > 0 \end{cases}$$



Data samples with **three** features (x_1, x_2, x_3)

The decision boundary is a **surface** for $z = 0$

The decision boundary can be shifted left or right via the bias

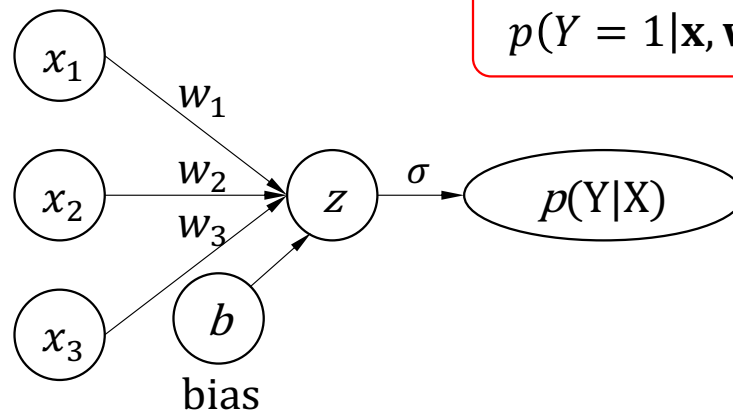
The decision boundary must cross the origin if no bias !

Logistic Regression

- (only need to) parameterise the $p(Y|X)$ without independence assumptions

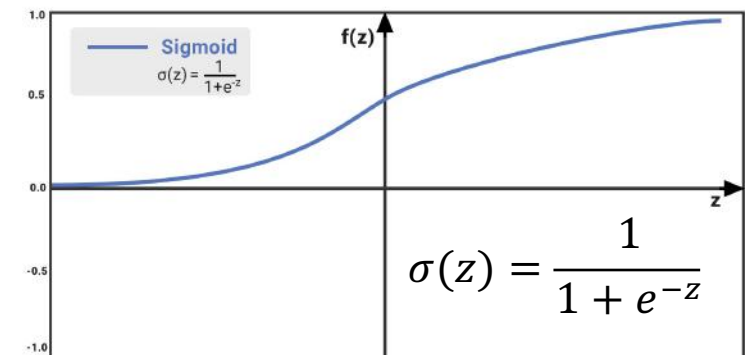
$$p(Y = 1|\mathbf{x}, \mathbf{w}, b) = f(\mathbf{x}, \mathbf{w}, b)$$

input layer output layer $z = \mathbf{w}^T \mathbf{x} + b$



$$p(Y = 1|\mathbf{x}, \mathbf{w}, b) = \sigma(\mathbf{w}^T \mathbf{x} + b)$$

Sigmoid/Logistic function



Logistic Regression

- Logistic regression does not require independence assumptions $x_i \perp \mathbf{x}_{-i}$, like Naïve Bayes
- Example, in spam classification, $x_1 = 1$ ["bank" exists] and $x_2 = 1$ ["account" exists]

If "bank" and "account" always appear together,

Naïve Bayes will count this evidence twice:

$$p(y, x_1, x_2, x_3) = p(y)p(x_1|y)p(x_2|y)p(x_3|y)$$
$$p(x_1|y) = p(x_2|y)$$

Logistic regressive can set either w_1 or w_2 to **zero** to ignore one of it!!

Logistic Regression

- Discriminative model is powerful, so what is the advantage of generative model?
 - Discriminative models $p(Y|X)$ require all X are observed,
fail to work if some inputs are missing!

- Generative models $p(Y|X) = \frac{p(Y,X)}{p(X)} = \frac{p(Y)p(X|Y)}{p(X)} \propto p(Y)p(X|Y)$

when some input are unobserved, still allow us to compute $p(Y|X)$

e.g., Naive Bayes

	$x_1 = 0$	$x_1 = 1$	$x_2 = 0$	$x_2 = 1$	$x_3 = 0$	$x_3 = 1$	$x_4 = 0$	$x_4 = 1$
$y = 0$	3	5	5	2	0	8	7	4
$y = 1$	1	0	3	10	7	4	2	5

- $p(Y = 0) = \frac{3+5+5+2+\cancel{8}+\cancel{7}+4}{(3+5+5+2+\cancel{8}+\cancel{7}+4)+(1+3+10+7+\cancel{4}+\cancel{2}+5)}$
- $p(x_1 = 0|Y = 0) = \frac{3}{3+5+5+2+\cancel{8}+\cancel{7}+4}$

- Problem of High-dimensional Data
- Less Parameters: Conditional Independence
- Less Parameters: Bayesian Network
- Naïve Bayes Classifier
- Discriminative vs. Generative Models
- Logistic Regression
- **Deep Neural Networks**
- Continuous Variables

Deep Neural Network

- Logistic regression parameterises the $p(Y|X)$ without independence assumptions

$$p(Y = 1|\mathbf{x}, \mathbf{w}, b) = f(\mathbf{x}, \mathbf{w}, b)$$

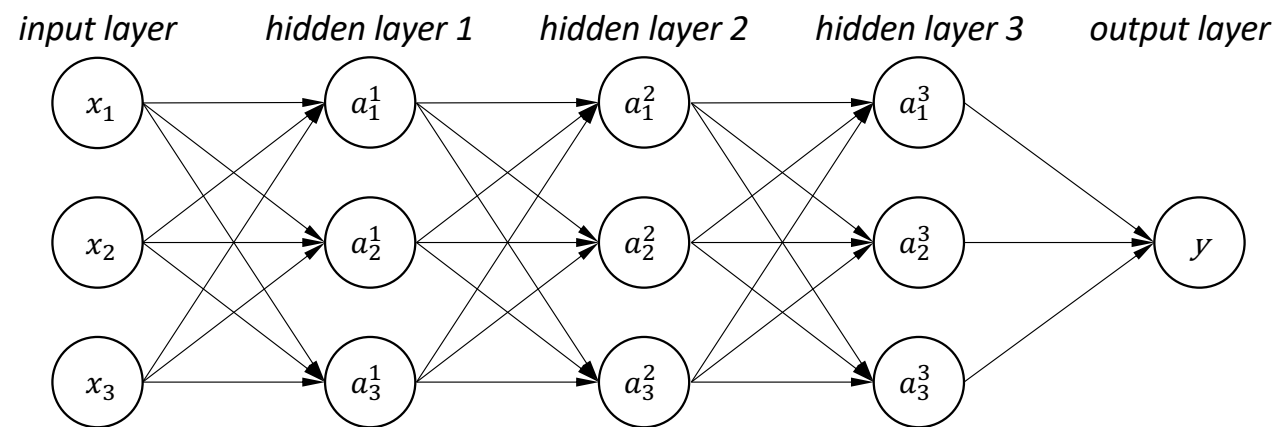
but logistic regression is a **linear dependence** (between input and output)
which might be too simple

Non-linear dependence is better ...

$$p_{\text{Neural}}(Y = 1|\mathbf{x}, \boldsymbol{\theta}) = f(\mathbf{x}, \boldsymbol{\theta})$$

Deep Neural Network

More parameters and layers, better representation capacity ...



More powerful than logistic regression

Deep Neural Network

- Naïve Bayes

$$\begin{aligned} p(x_1, x_2, x_3, x_4) &= p(x_1)p(x_2|x_1)p(x_3|x_1, x_2)p(x_4|x_1, x_2, x_3) \\ &\approx p(x_1)p(x_2|x_1)p(x_3|\cancel{x_1}, x_2)p(x_4|\cancel{x_1}, \cancel{x_2}, x_3) \\ &\approx p(x_1)p(x_2|x_1)p(x_3|x_2)p(x_4|x_3) \end{aligned}$$

- Deep Neural Network

$$\begin{aligned} p(x_1, x_2, x_3, x_4) &= p(x_1)p(x_2|x_1)p(x_3|x_1, x_2)p(x_4|x_1, x_2, x_3) \\ &\approx p(x_1)p(x_2|x_1)p_{\text{Neural}}(x_3|x_1, x_2)p_{\text{Neu}ral}(x_4|x_1, x_2, x_3) \end{aligned}$$

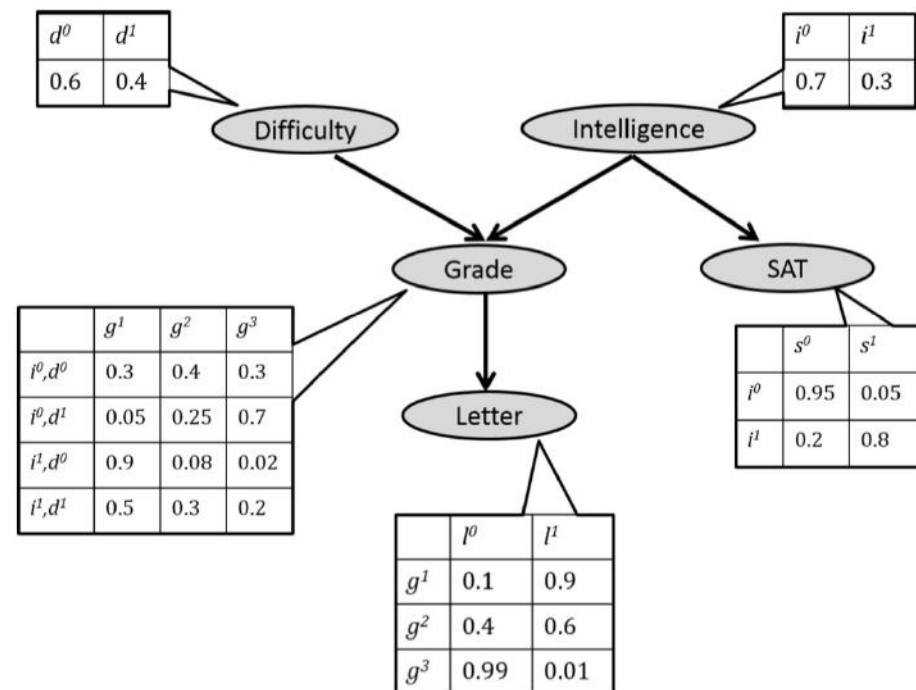
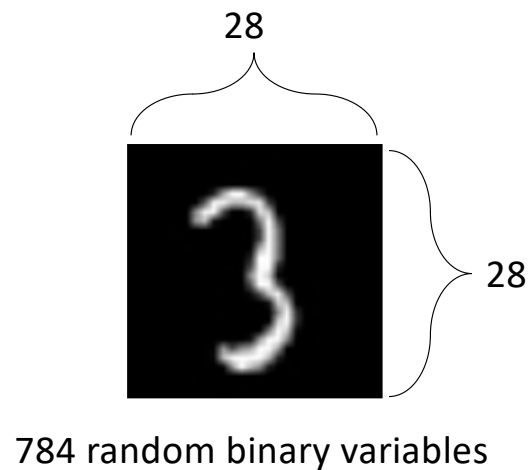
- Problem of High-dimensional Data
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- **Continuous Variables**

Continuous Variables

- Discrete Variables

The below examples both use discrete variables, but there are many variables are continuous!

e.g., age, height ...



Continuous Variables

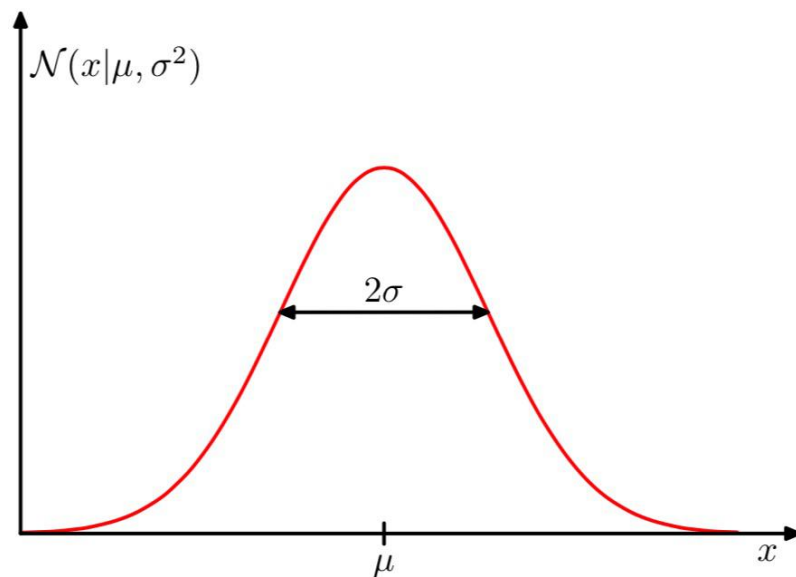
- **Represent Continuous Variables**

If x is a continuous variable, we can represent it with its **probability density function (PDF)** instead of a **table** anymore ..

Continuous Variables

- Represent Continuous Variables

Consider x is a random **float-point** variable to represent “age”,
we can use 1-D Gaussian to parameterised the density.



$$\mathcal{N}(x|\mu, \sigma^2) = \frac{1}{(2\pi\sigma^2)^{1/2}} \exp \left\{ -\frac{1}{2\sigma^2} (x - \mu)^2 \right\}$$

$$\mathcal{N}(x|\mu, \sigma^2) > 0$$

μ : mean

σ : standard deviation

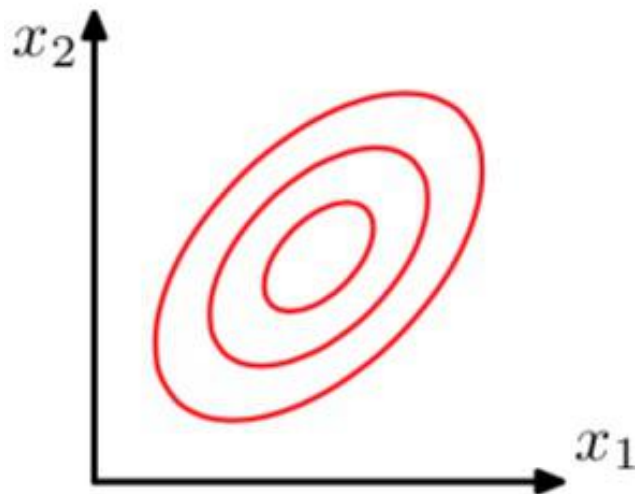
σ^2 : variance

$\beta = \frac{1}{\sigma^2}$: precision

Continuous Variables

- Represent Continuous Variables

Consider $\mathbf{x}$ is a random **float-point** vector to represent “age”, “height”, “weight”
it can be a **joint probability density function**
we can use **D-dimensional Gaussian** to parameterize it
(a.k.a Multivariable Gaussian)



$$\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{D/2}} \frac{1}{|\boldsymbol{\Sigma}|^{1/2}} \exp \left\{ -\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}) \right\}$$

$\boldsymbol{\mu}$ is called the mean

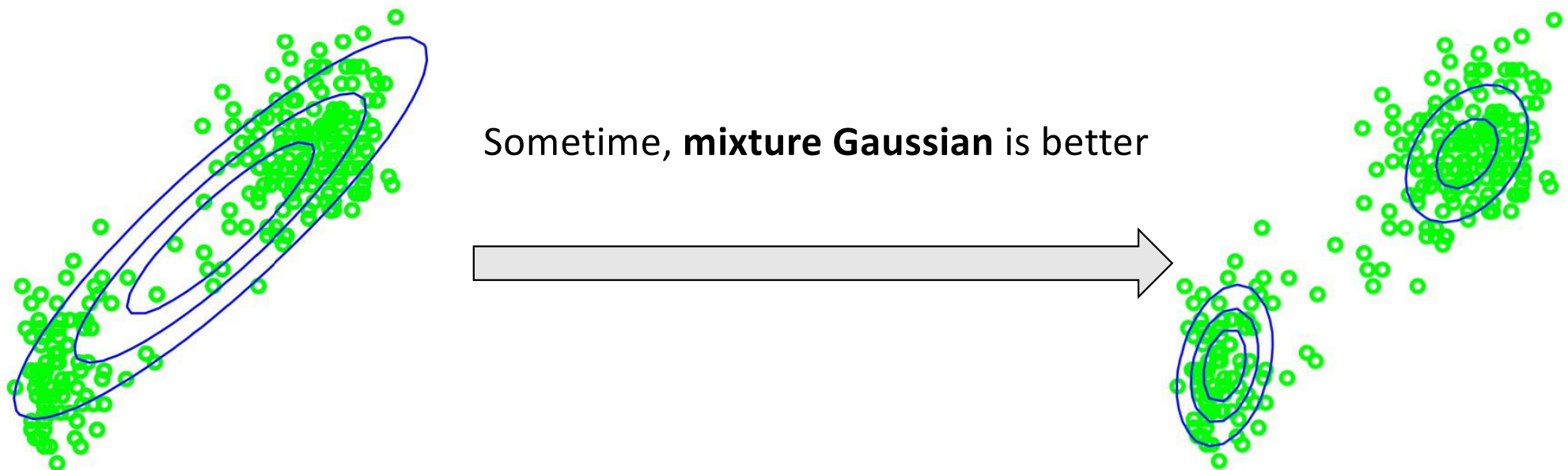
$\boldsymbol{\Sigma}$ is called the covariance

$|\boldsymbol{\Sigma}|$ denotes the determinant of $\boldsymbol{\Sigma}$

Continuous Variables

- Represent Continuous Variables

Consider $\mathbf{x}$ is a random **float-point** vector to represent “age”, “height”, “weight”



Data Representation

- How to do representation
 - Problem of High-dimensional Data
 - Less Parameters: Conditional Independence
 - Less Parameters: Bayesian Network
- How to do inference
 - Naïve Bayes Classifier
 - Discriminative vs. Generative Models
 - Logistic Regression
- How to be better
 - Deep Neural Networks
 - Continuous Variables

Thanks